Introducción a la Electrónica

Unidad 5

Dispositivos de Disparo

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El Tiristor

Segunda Parte

(Rectificador Controlado de Silicio o SCR)

Dr. Ing. David M. Petruzzi

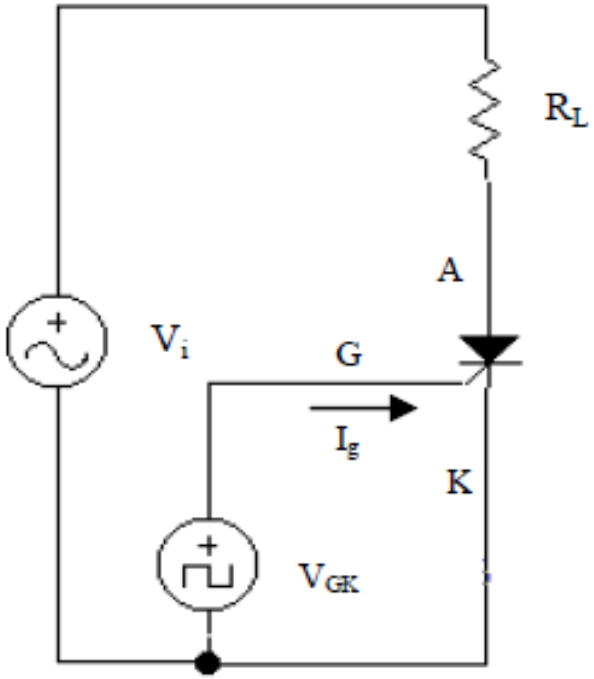
Disparo aplicando pulsos de corriente en Compuerta

5.1.5 – Circuito de Prueba :

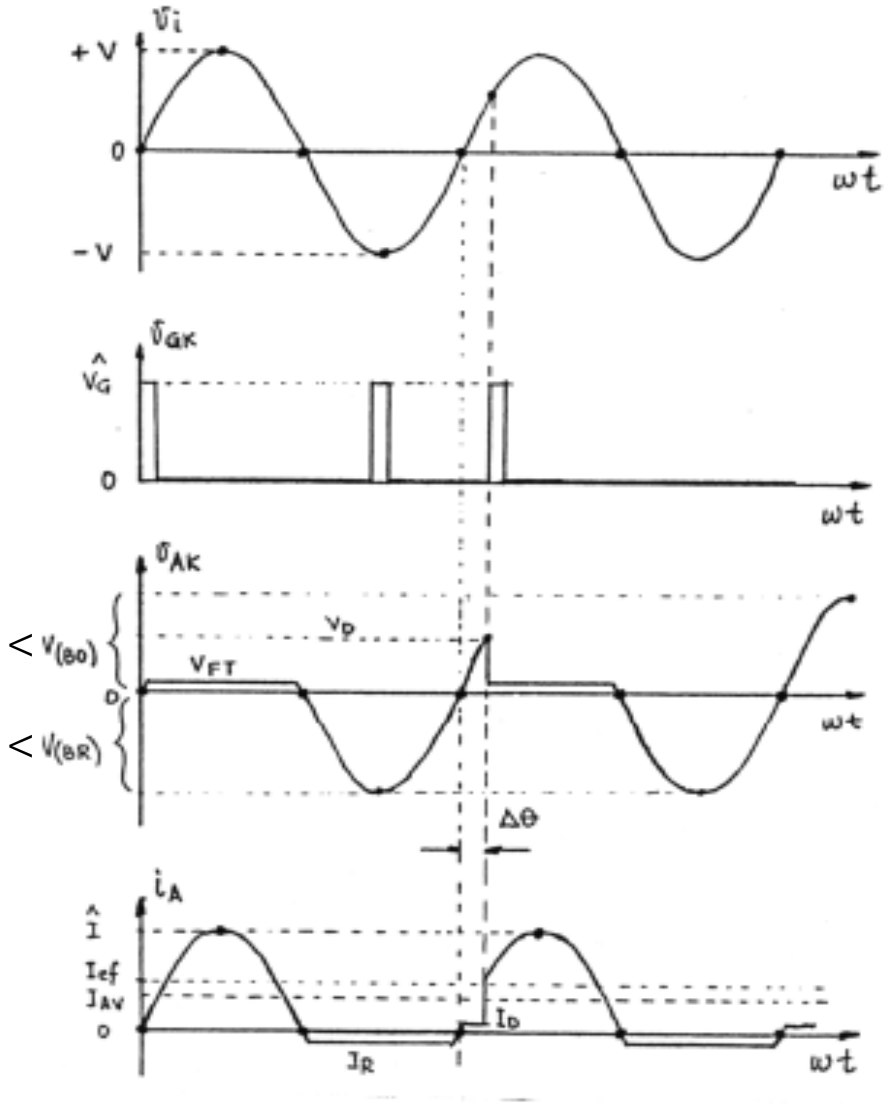
$v_i = V \cdot \text{sen}\ \omega.t = V \cdot \text{sen}\ \theta$

$V < V_{(BR)R}$ (ruptura inversa)

$V_{RL} = v_i - v_{AK} = i_A \cdot R_L$



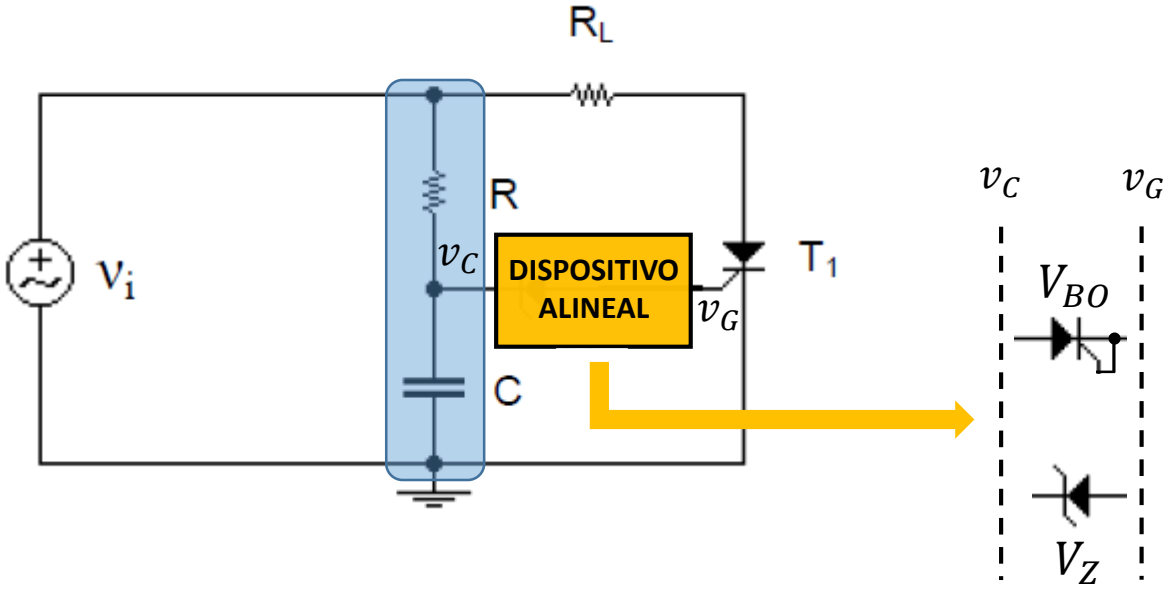
Formas de onda :



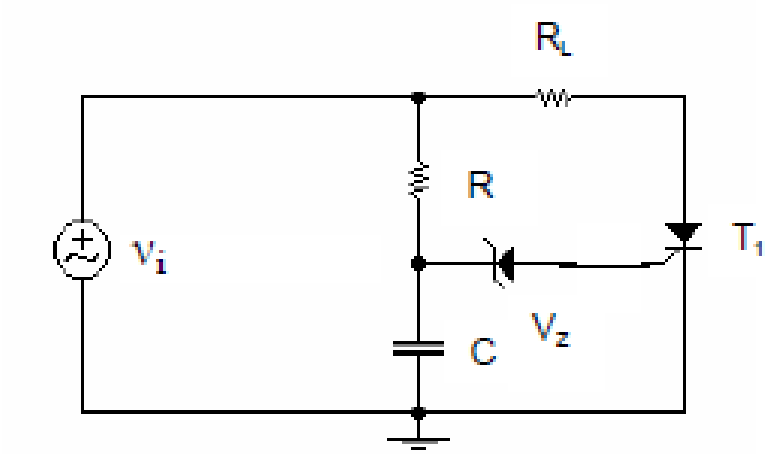
Disparo empleando

RC pasabajos + Dispositivo Alineal

en Compuerta



En base al circuito mostrado a continuación:



R = 1 kΩ

C = 3 μF

$v_i = 170V.\ sen\ [(100.\pi).t]$
$R_L = 10\ \Omega$
$V_Z = 50\ V$
$V\gamma = 0{,}6\ V$

<u>Datos T_1:</u>
$V_{BO} = 200\ V$; $V_R = 300\ V$
$I_L = I_H = 0$
$dV/dt\ máx = 10^6\ V/seg$

1. Hallar y representar las formas de onda de tensión en:
   a) el capacitor
   b) el tiristor
   c) la carga

2. Calcular, en base a las mismas:
   a) el ángulo de demora
   b) el ángulo de conducción
   c) la potencia en la carga

1ro. Analicemos solamente la malla RC con el generador de CA

$$v_i(t) = V_i . sen\ \omega t$$

$$\dot{V}_i = V_i \underline{|0}$$

$$I_1(j\omega) = \frac{V_i(j\omega)}{Z(j\omega)} = \frac{V_i(j\omega)}{R + \frac{1}{j\omega C}}$$

$$\dot{I}_1 = I_1 \underline{|\varphi}$$

$$I_1 = \frac{V_i.(\omega C)}{\sqrt{1 + (\omega RC)^2}}$$

$$\varphi = \left(\frac{\pi}{2}\right) - tg^{-1}(\omega RC)$$

$$V_C(j\omega) = I_1(j\omega).\frac{1}{j\omega C} = \frac{V_i(j\omega)}{j\omega RC + 1} = -j\ \frac{I_1(j\omega)}{\omega C}$$

$$\dot{V}_C = \frac{I_1}{\omega C}\underline{|[\varphi - (\pi/2)]} = \frac{V_i}{\sqrt{1 + (\omega RC)^2}}\ \underline{|-\theta_i}$$

$$\theta_i = (\pi/2) - \varphi = (\pi/2) - (\pi/2) + tg^{-1}(\omega RC) = tg^{-1}(\omega RC)$$

$$\theta_i + \varphi = \pi/2$$

$$V_R(j\omega) = I_1(j\omega).R = \frac{V_i(j\omega)}{R + \frac{1}{j\omega C}}.R = \frac{V_i(j\omega)}{j\omega RC + 1}.j\omega RC = j\ V_C(j\omega).(\omega RC)$$

$$\dot{V}_R = (I_1.R)\underline{|\varphi}$$

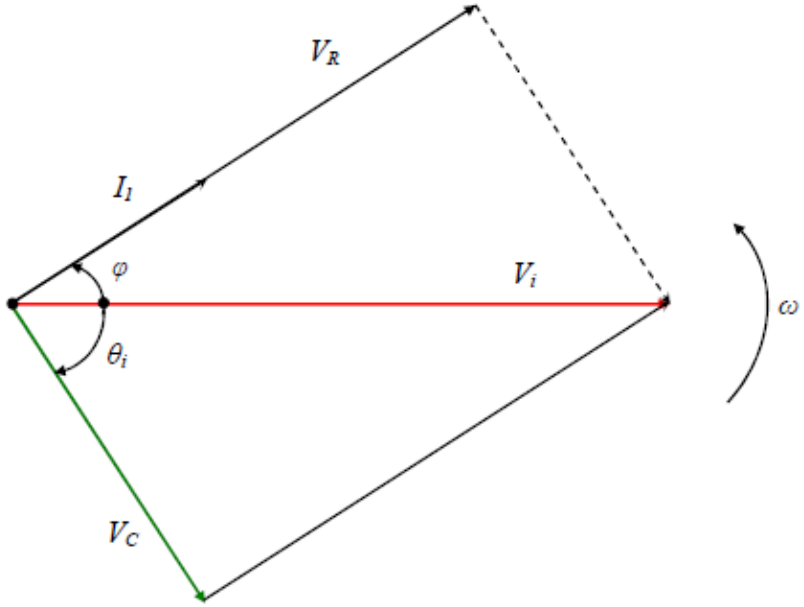
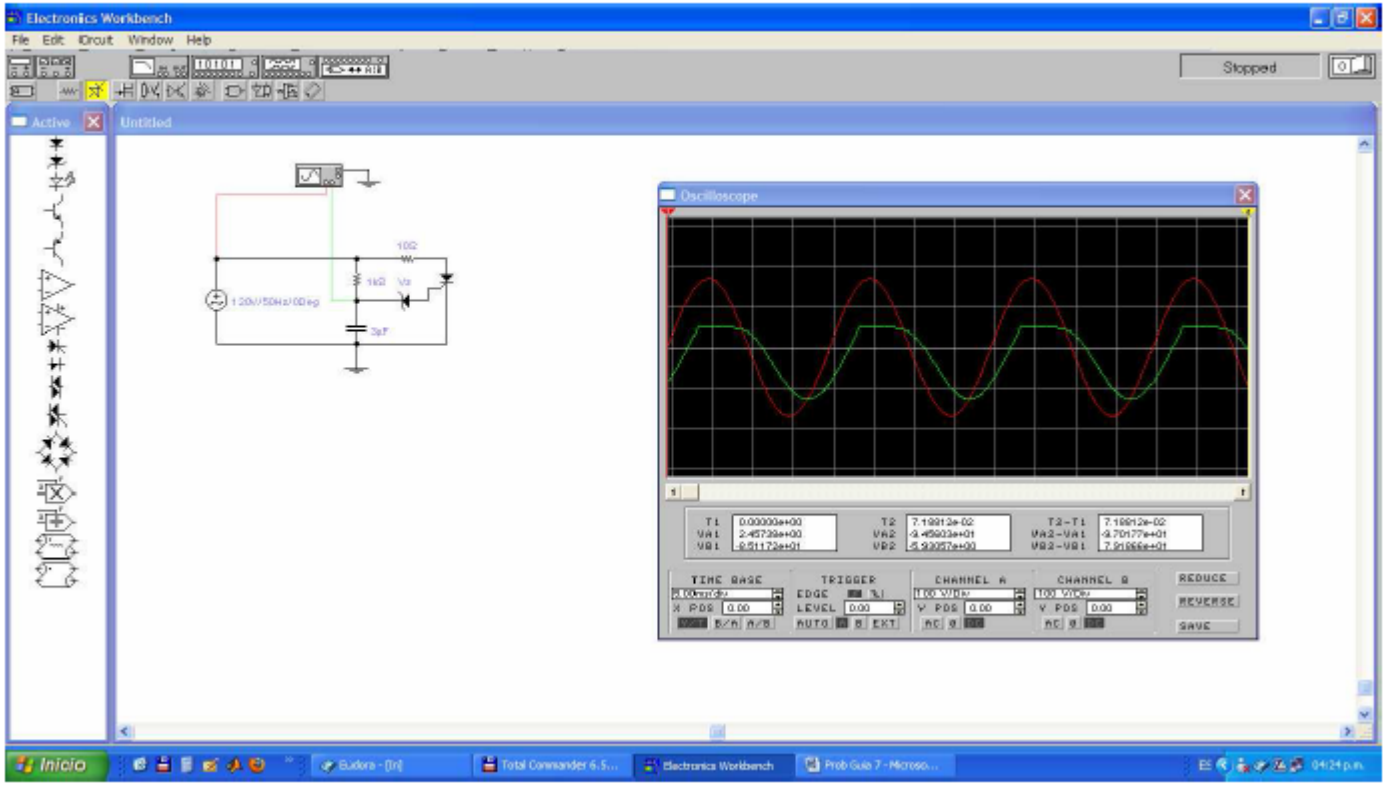
$$v_C(t_i) = V_C.sen(\omega t_i - \theta_i) = V_C.sen(\alpha) = V_Z$$

$$\alpha = sen^{-1}\left(\frac{V_Z}{V_C}\right)$$

$$\theta_d = \theta_i + \alpha$$

$$\theta_c = \pi - \theta_d = \pi - (\theta_i + \alpha)$$

2do. Efectuemos el Diagrama Fasorial correspondiente:



3ro. Simulando el circuito con el EWB, se obtiene: